Modelling NFL Statistics to Create a Virtual Interactive Gambling Model

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1. Introduction:

The primary objective of this project is to develop an interactive betting model capable of predicting various metrics in the National Football League (NFL). Although the dataset is versatile and can be adjusted to forecast a range of variables, this project will specifically focus on predicting the final scores of NFL games. This endeavor involves leveraging advanced statistical techniques and machine learning algorithms, specifically linear regression, to analyze historical game data and make accurate predictions, ultimately aiding in betting decisions.

2. The Data:

2.1 Description of the Data and Significant Variables

The Data contains different variables that encompass the game of American Football and specifically the NFL. The Data was gathered by combining statistics from both the ESPN website and Pro-Football-Reference.com to ensure all variables that drive NFL games were included. Before I describe the variables, here's a quick synopsis of the rules of the games

Overview of the NFL Rules and Scoring System:

The NFL, established in 1920, is the professional American football league in the United States, comprising 32 teams divided into two conferences: the American Football Conference (AFC) and the National Football Conference (NFC). Each NFL game is played over four quarters, each lasting 15 minutes, with a halftime break after the second quarter. If the game is tied at the end of regulation, it proceeds to overtime, which has specific rules to determine the winner.

Scoring in the NFL:

Points in an NFL game can be scored in several ways:

- Touchdown (6 points): Achieved when a team carries or catches the ball in the opponent's end zone.
- Extra Point (1 point): Attempted after a touchdown, where the team kicks the ball through the opponent's goalposts from the 15-yard line.
- Two-Point Conversion (2 points): Instead of an extra point, a team can attempt to get the ball into the end zone again from the two-yard line.
- Field Goal (3 points): Scored by kicking the ball through the opponent's goalposts during a play from scrimmage.
- Safety (2 points): Occurs when the offensive team is tackled in their own end zone or commits certain fouls in their end zone.

The Downs System:

The downs system is a fundamental aspect of NFL gameplay, dictating how the offensive team advances the ball and maintains possession. The offense has four downs (plays) to move the ball at least 10 yards down the field. If they succeed, they earn a new set of four downs. If they fail to gain 10 yards after four downs, possession of the ball is turned over to the opposing team.

- First Down: The initial attempt to advance the ball. If the team gains 10 or more yards, they receive a new set of downs.
- Second Down: The second attempt if the team did not gain 10 yards on the first down.
- Third Down: The third attempt, which often determines whether the team will attempt a riskier play to gain the remaining yards.
- Fourth Down: The final attempt. If the team is close to the required 10 yards, they might attempt to convert. Otherwise, they typically punt the ball to the opposing team or attempt a field goal if within range.

The Role of the Quarterback:

In American football, the quarterback (QB) is one of the most critical positions on the field. The quarterback is often seen as the leader of the offense, responsible for directing plays, making strategic decisions, and executing passes. The QB stands behind the offensive line and receives the ball from the center to start each play. Their primary duties include:

- Passing: Throwing the ball to receivers, running backs, or tight ends to advance down the field.
- Rushing: Occasionally running with the ball themselves to gain yardage.
- Play Calling: Making quick decisions on which play to execute based on the defensive setup.

Understanding these rules and the scoring system is crucial for building an accurate predictive model, as they form the basis for how game outcomes are determined. This project will utilize this foundational knowledge to analyze game data and develop a model capable of predicting scores, enhancing the ability to make informed betting decisions.

Variables:

Key:

To avoid explaining every variable in the data I will explain the 10 significant variables that will be selected in our model through careful consideration.

- 'Score' = This is the dependent variable. It is the Score accumulated during the game.
- 'Total Drives' = This denotes each possession a team gets. When one's teams Offence is on the field, the opposing teams Defense is present on the field to try and stop the opposition scoring. A 'Drive' denotes an offensive possession.
- 'Passing Yards' = Quite simply this is the total number of yards thrown by the quarterback during the game.
- 'Rushing' = Similar to passing yards this variable represents the numbers of yards gained running with the ball in your hands rather than throwing it.
- 'Turnovers' = This variable denotes the number of times a team gives away possession. It always comes as a mistake. Turnovers include interceptions and fumbles which are both included in our data frame but will be removed due to them summing up to 'Turnovers'
- 'Sacks' = This denotes the number of times the quarterback gets tackled while he is in possession of the ball resulting in a loss of a down.
- 'Punts' = To punt is to kick the ball back to the other team. By kicking it down the field you force them to start their possession or 'Drive' as far away from your goal as possible. Teams punt when they run out of downs.
- '3rd downs Conv' = A team gets a total of four goes to get 10 yards. when it's their 3rd turn, it's called a third down. If you don't complete it, you have to kick the ball down the field a 'Punt' or you can opt to go for it on a 4th turn called a 4th down. If you fail to make the 4th down the opposing team takes over where you attempted the play. The 3rd down converted variables indicate how many of these successful 3rd downs a team has.
- '3rd downs ATT' = Similarly this is the number of attempts at 3rd downs.
- '4th downs ATT' = the number of 4th downs a team attempts.

- '4th downs Conv' = indicates the number of 4th downs converted by a team in a game. Playing the ball on 4th down is risky due to the threat of the opposing team getting the ball close to your goal.

These are the variables that will be selected in our final model.

2.2 Data Cleaning:

The Data was gathered by combining statistics from both the ESPN website and Pro-Football-Reference.com to ensure all variables that drive NFL games were included. The data contains no blanks. The dataset comprises 570 observations, representing a total of 285 games for the season, including the post-season. Among these, 14 games proceeded to overtime. Overtime occurs when the teams are tied at the end of the regulation period, resulting in additional playtime.

Due to the extra time allocated to games that go into overtime, these games have additional data associated with them. This extra data could introduce bias into the model, making it difficult to assess the games fairly. To ensure the integrity of the model, I will remove the observations related to overtime games from the dataset. The removal of these 14 games will result in the exclusion of 28 observations from the season's data.

Additionally, the Total Passing Yards statistic was standardized. In football, the total passing yards given typically includes the yards lost due to sacks. A sack occurs when the quarterback is tackled behind the line of scrimmage before they can throw a forward pass, resulting in a loss of yardage. To standardize the Passing Yards column and provide a more accurate measure of a team's passing performance, the yardage lost due to sacks was added back to the Passing Yards column. This adjustment ensures that the passing yardage reflects only the yards gained through passing plays, without penalizing teams for sack yardage lost. This standardization is crucial for maintaining consistency and accuracy in the dataset, thereby enhancing the reliability of the predictive model.

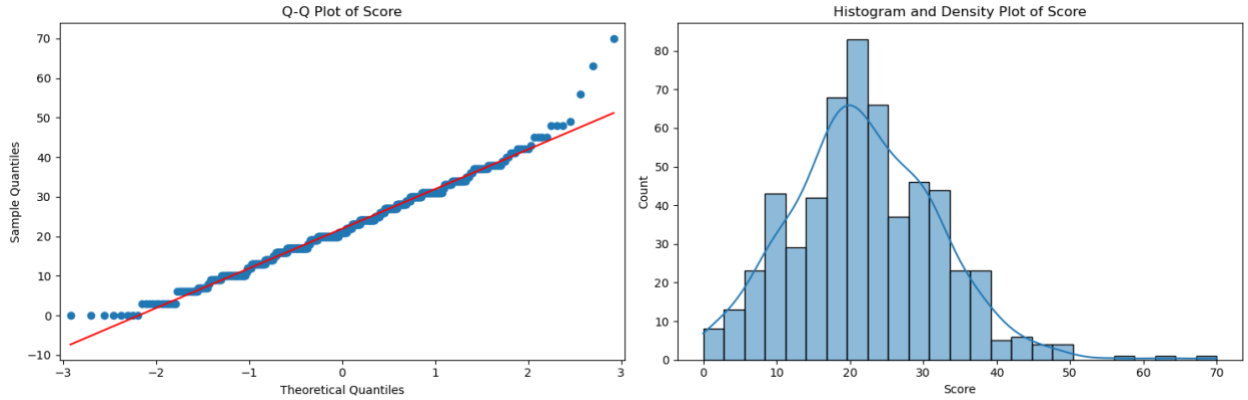
The variable 'Passer Rating' is calculated using various other variable as shown here. This variable is removed as it clearly will introduce multicollinearity into the model,

$$Passer\ Rating = \left(\frac{\left(\frac{COMP}{ATT} - 0.3\right) * 5 + \left(\frac{YARDS}{ATT} - 3\right) * 0.25 + \left(\frac{TD}{ATT}\right) * 20 + 2.375 - \left(\frac{INT}{ATT} * 25\right)}{6} \right) * 100$$

We can see that this variable emcompasses data from 5 different variables. The most worrying the the 'TD' variable which indicates a touchdown in the game. We will try to avoid scoreing variables as will be discussed later as they completely drive the model.

3. Exploratory Analysis:

3.1 Dependent Variable:



The provided plots analyze the dependent variable, which in this case is the score. The two plots are:

- Q-Q Plot of Score
- Histogram and Density Plot of Score

Q-Q Plot of Score

The Q-Q (Quantile-Quantile) plot compares the distribution of the scores to a theoretical normal distribution.

Interpretation:

- Most points lie along the red reference line, indicating that the scores generally follow a normal distribution.
- Deviations at the tails suggest the presence of outliers, with higher-than-expected scores evident in the upper right corner.
- Lower-end deviations are less pronounced but still present.
- Given the nature of NFL games, outliers are expected due to the unpredictable performance of players, particularly quarterbacks.

Histogram and Density Plot of Score.

The histogram shows the frequency distribution of scores, while the density plot provides a smoothed estimate.

Interpretation:

- The distribution is roughly bell-shaped, peaking around the 20-25 score range.
- There is a right skewness with a long tail extending towards higher scores, indicating outliers.

These outliers, resulting from exceptional performances, are typical in sports data.

The Q-Q plot and histogram reveal that while scores generally follow a normal distribution, there are expected outliers and slight right skewness. These outliers reflect the inherent unpredictability of NFL games and will not be handled separately, reinforcing the suitability of linear regression for predicting game scores.

3.2 Removal of Scoring Variables that Drive the Model.

In developing a robust predictive model, it is essential to carefully select the variables included in the analysis. For this project, I will remove variables that directly involve scoring, as their inclusion would drive the model's performance in an obvious and uninformative manner. Including these variables would lead to a tautological situation where the model simply identifies the direct correlation between scoring variables and the final score, rather than uncovering more nuanced predictors.

Effect of scoring variables in the model:

Passer Rating	−0.0150	0.018	−0.717	0.474	−0.049	0.023
Rush TDs	5.1232	0.324	15.813	0.000	4.487	5.760
FGM	2.7189	0.223	12.169	0.000	2.280	3.158
FGA	−0.5632	0.242	−2.332	0.020	−1.038	−0.089
XPM	0.8783	0.357	2.457	0.014	0.176	1.580
XPA	−0.5023	0.415	−1.209	0.227	−1.318	0.314
Punts	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

We can already see from this regression summary that the variables that directly involve scoreing completely drive the model.

- Defensive/Special Teams TDs: Defensive/special teams touchdowns
- Pass TDs: Passing touchdowns
- Rush TDs: Rushing touchdowns
- XPM: Extra points made
- XPA: Extra points attempted
- FGM: Field goals made
- FGA: Field goals attempted
- RZ ATT: Red zone attempts
- RZ Conv: Red zone conversions

By removing these scoring variables, the model will be challenged to identify other influential factors that contribute to the final game scores, such as team performance metrics, player statistics, and situational factors. This approach enhances the model's ability to generalize from the data and improves its utility for making predictions in scenarios where direct scoring information is not available. The goal is to build a model that relies on indirect indicators and broader performance metrics, thereby offering more strategic insights for predicting NFL game outcomes.

We can tell the model is currently depending on these variables based on the coefficients for these metrics. Notice tat the coefficient for 'FGM' which indicates Field Goals Made is 2.7189, this is nearly exactly equal to the value of a field goal which is three points.

3.3 Weather Variable

To understand the impact of weather on NFL game scores, I created a composite weather variable by combining 'Wind' and 'Weather' data. This new variable was split into categories,

with the intention of capturing different weather conditions that could affect game outcomes. The categories were:

- Clear, cloudy, overcast
- Rain
- Snow
- Windy (wind speeds greater than 13 mph)
- Windy with either snow or rain

Initially, I used one-hot encoding to create dummy variables for these categories. However, the forward selection model identified only the dummy variable for rain as significant, and even then, not all the time. It depended on what variables were included for selection. This seemed counterintuitive. To gain better insight, I calculated the average scores for each weather category:

Overall average: 21.9 points per game (ppg)

Clear, cloudy, overcast: 22.56 ppg

Rain: 19.275 ppg

Snow: 17.1 ppg (3 observations)

Wind: 18.869 ppg

Wind with rain or snow: 17.1 ppg

These averages indicated clear differences in game scores based on weather conditions. Despite these differences, the one-hot encoded variables did not perform well in the regression model, prompting me to consider an ordinal encoding approach. The ordinal coding ranked the categories from highest to lowest average score:

Clear, cloudy, overcast = 0

Rain = 1

Wind = 3

Wind with rain or snow = 4

Snow = 5

I then tested models with various binary splits for weather conditions, such as:

- Wind binary: 0 = clear, 1 = rain or snow
- Weather binary: 0 = clear, cloudy, overcast, rain; 1 = snow, wind, windy with rain or snow

Evidence of these tests on the Weather variable can be found in the Pythin environment attached with this project.

Despite these efforts, the weather variables still did not emerge as significant predictors in the forward selection regression model. One possible explanation for this outcome is the increasing number of NFL games played in indoor stadiums. Currently, 11 NFL teams play their home games indoors, with each team playing roughly 8-9 home games per season. This results in

nearly 100 observations of games played indoors, reducing the variability and impact of weather conditions in the dataset. Additionally, there were only three observations of snow over the 285 games, limiting the statistical power to detect significant effects.

While the weather variable was not ultimately included in the final model, the calculated averages suggest that with more observations, especially of adverse weather conditions, the impact of weather on game scores could become significant. This insight highlights the importance of considering environmental factors in sports analytics and the need for larger datasets to capture their effects accurately. The problem with the NFL and viewing historical data for team to train regression is the huge turnover on rosters. On average an NFL roster turn over 25% of its players every year according to “ESPN”. The weather data used in this analysis was sourced from NFLWeather.com, which provided comprehensive information on game-day weather.

Momentum Variable:

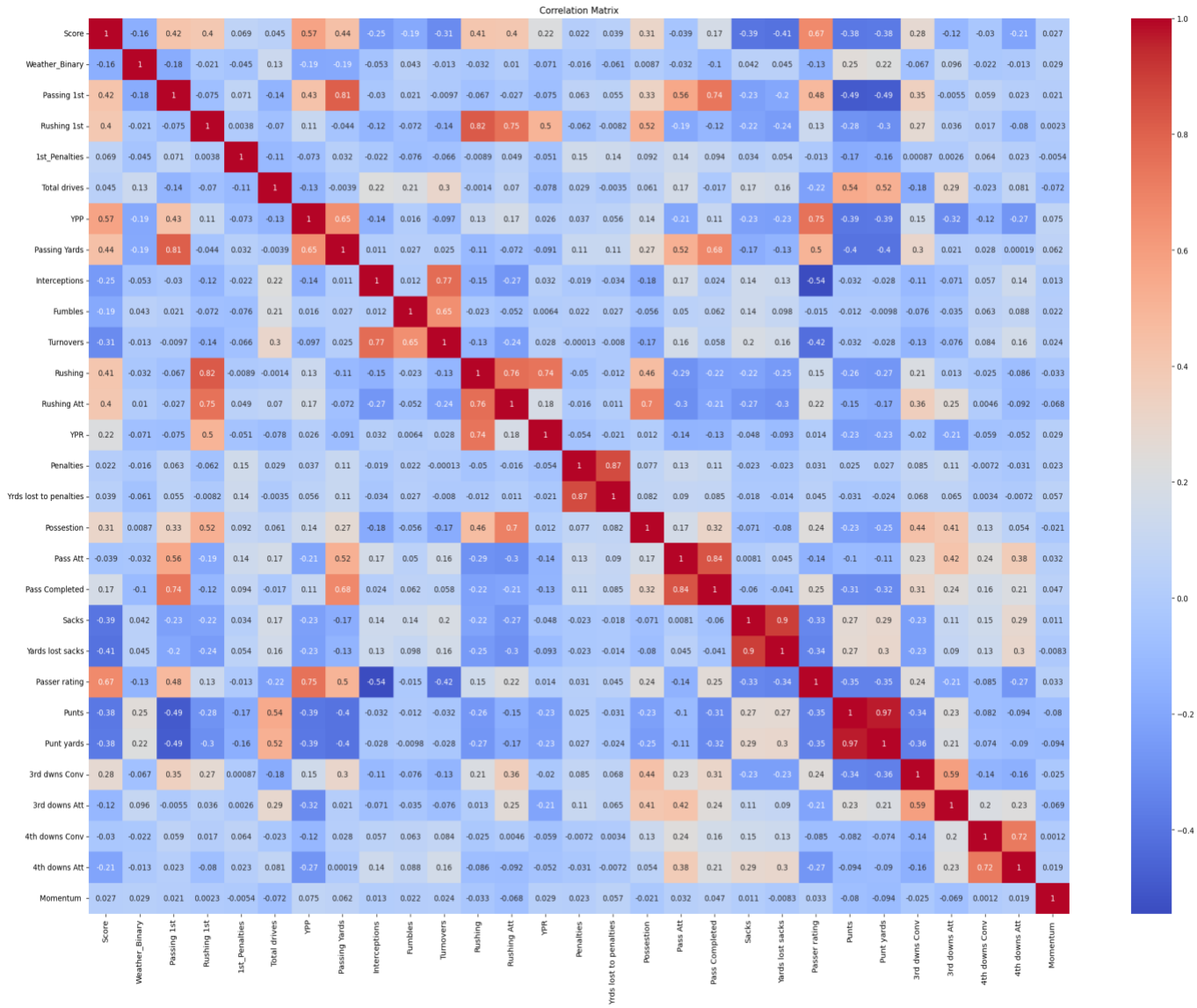
During the project, I initially attempted to create a variable to monitor the momentum of teams following wins and losses. The idea was to capture the psychological and performance impacts of previous game outcomes on subsequent games. However, as the project progressed, it became clear that the method of training the regression model naturally accounted for momentum effects.

The regression model was trained iteratively, week by week, incorporating each team's performance over time. This approach implicitly captured the momentum of teams. If a team performed poorly in a given week, this performance was incorporated into the model, thereby lowering their averages and affecting subsequent predictions. Conversely, if a team performed well, their improved performance was also reflected in the model, raising their averages and influencing future predictions.

This realization highlighted that the week-by-week training method effectively captured the dynamic changes in team performance, rendering the creation of the additional momentum variable redundant. This approach ensured that the model remained responsive to the latest team performance trends, providing a more accurate and timely reflection of a team's form and momentum.

3.4 Correlation Matrixes

It is only natural that a dataset such as this would show evidence of multicollinearity. To address this issue, I will use a combination of techniques, including analyzing correlation matrices, employing forward selection, and evaluating the Variance Inflation Factor (VIF) later when evaluating the model. After examining the correlation matrix, I will also review scatter plots and histograms to ensure that variables I have selected for the forward regression show some sort of linear relationship with the dependent variable



--We can see that '4th downs Conv' variables is highly correlated with '4th downs Att'. This is the same for the variables '3rd downs Conv' and '3rd downs Att'. I will leave these for now as I don't know which seems to be more important. Is it more important to have more attempts or to be efficient? It's possible both are significant; I would expect to see a negative coefficient for attempts and positive coefficient for converted.

--'Punts' and 'Punt yards' are highly correlated. I will remove the variable for 'Punt yards'. The important factor here is that when punting you are giving the ball away to the other team. The distance punted is unimportant compared to act of losing the ball.

--'Sacks' and 'Yards lost sacks' are highly correlated. I will eliminate 'Yards lost sacks' as I assume the model will average out the number of 'Yards lost sacks' for each sack. The important thing here is that when sacked the team loses a 'down' the team gets 4 downs to get 10 yards to

retain the ball. therefore, there is more importance to being sacked than for what distance you get sacked, even though the distance is important.

--'Rushing', which indicates total rushing yards is highly correlated with 'Rushing 1st', 'YPR' and 'Rushing attempted'. I want to delete 'Rushing attempted', 'YPR' as its more important the number of Rush yards obtained. But I want the model to capture %plays passed and %plays rushed. Some teams pass and rush more than others. I think the total yards rushed will explain YPP and rushing attempted as the model can distinguish which teams rush the ball more based on their total rushing yards.

--'Passing Yards' is highly correlated with 'Passing 1st', 'YPP' (yards per pass), 'Pass ATT', Pass Completed'. It is understandable that these are highly correlated as they all keep track of the same metric. It is interesting to see that 'Passing Yards' is highly negatively correlated with 'Punts' and 'Punt yards', this makes sense that the more you punt the ball the less passing yards you may achieve due to the act of giving possession away. 'Passing Yards' and 'YPP' are highly correlated with the dependent variable .44 and .57 respectively, this led me to cautiously follow the same steps as I did with the rushing variable as YPP is more highly correlated with the dependent variable than 'Passing Yards'. I will remove 'YPP' as in dummy selection trials it was never selected, and 'Passing Yards' was. I will remove 'Pass ATT', 'Pass completed' and 'YPP'

--'Penalties' and 'Yrds lost to penalties' are highly correlated. Unlike the Sack column, when a penalty is committed, they don't lose a turn they just lose yards. Either way I will remove the 'Yrds lost to penalties' variable with the understanding the model will be able to capture the average yards lost per penalty from the 'Penalties' Column. Both variables are not correlated with the dependent variable. Also, 1st downs from penalties are the real winner of these variables. This indicates when a team gets a first down because of a mistake made by the other team.

--I will remove the variables 'Fumbles' and 'Interceptions'; they sum to 'Turnovers' in the data.

--I will remove 'Rushing 1st' and 'Passing 1st' from this data. The number of first downs you get is important. When you get a first down you retain the ball. but the object is to get the ball down the field and score. You can do this with 1 down or 20 downs. Therefore, I don't feel this helps the model. They are also extremely highly correlated with their respected other 'Rushing' and 'Passing' predictors.

-- I concluded that 'Possession' should be removed from the model due to how correlated it is with other variables. Its data is captured by other variables such as total drives and total passing etc. Also, the time of possession seems to be reasonably centered indicating teams normally share the distribution of time on the ball. Another issue for the variable 'Possession' is that I would have expected a positive relationship. The longer you have the ball the more opportunities you must score. This is not reflected in the original model with its negative coefficient. One can only assume that efficiency far outweighs time of possession.

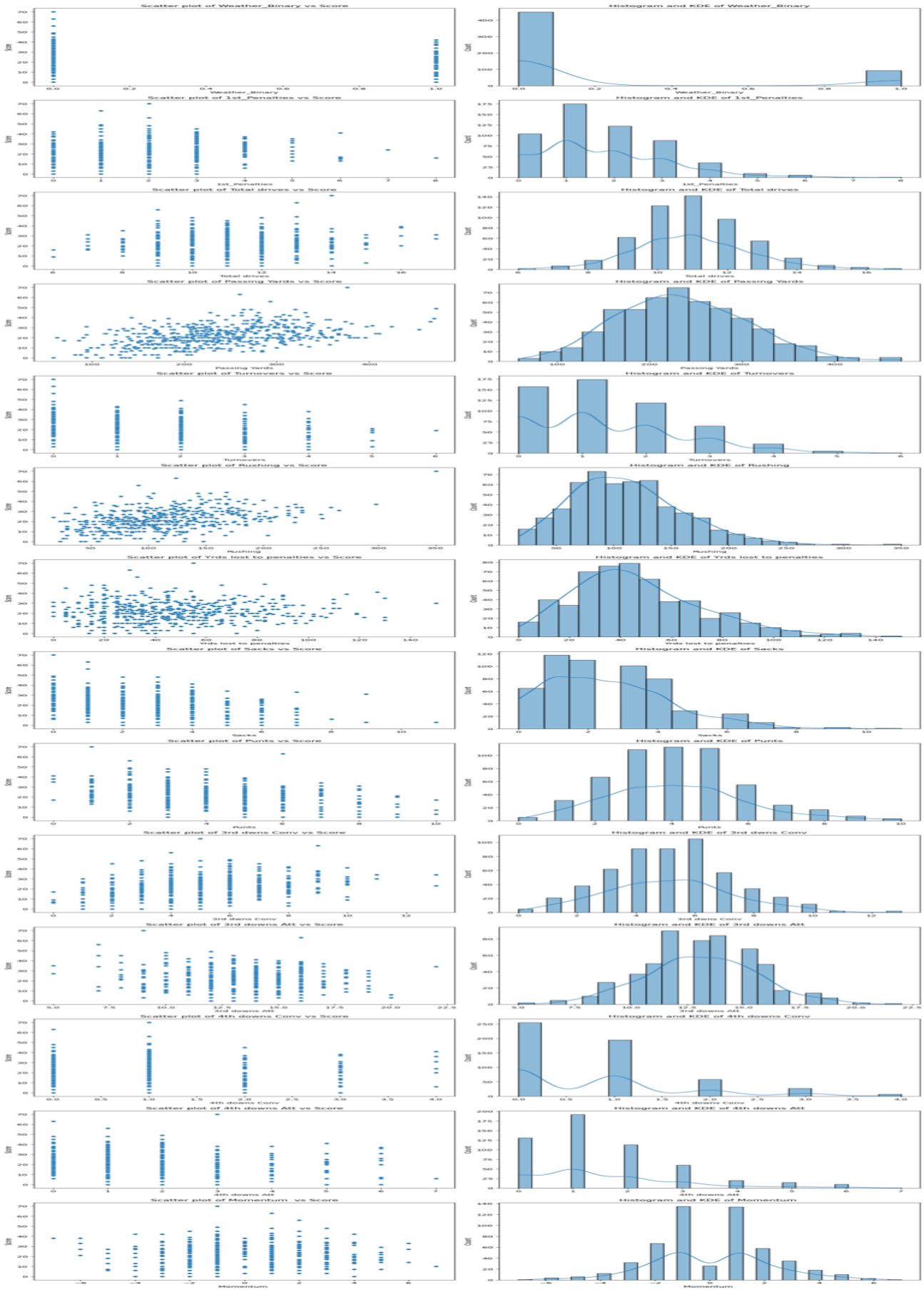
--'Temp' is removed as its data should be captured by 'Weather_Binary', it's also not correlated with the dependent variable.

--Although an important metric for evaluating Quarterback performance, and its high correlation with the dependent variable (.67) the 'Passer Rating' variable will be removed. It is calculated using a formula that involves several other variables, so its multicollinearity is expected to be very high.

3.5 Scatterplots and Distributions:

Before removal I will check the plots of the predictor variables,

We can see below that most of the scatterplots follow a positive or linear trend, most of the distributions are approximately normal with evidence of right skewness. This is to be expected when modelling sports scores. Most games will stay around the means but exceptional performance in these metrics drives the skewness. It’s interesting to see that 'Weather_Binary' seems to have a weak negative trend suggesting bad weather leads to lower scores. None of the high scores are during bad weather. But I already know that it won’t be selected in my final model.


Scatter plot of Weather_Binary vs Score
Weather_Binary
Score
Histogram and KDE of Weather_Binary
Weather_Binary
Count
Scatter plot of 1st_Penalties vs Score
1st_Penalties
Score
Histogram and KDE of 1st_Penalties
1st_Penalties
Count
Scatter plot of Total drives vs Score
Total drives
Score
Histogram and KDE of Total drives
Total drives
Count
Scatter plot of Passing Yards vs Score
Passing Yards
Score
Histogram and KDE of Passing Yards
Passing Yards
Count
Scatter plot of Turnovers vs Score
Turnovers
Score
Histogram and KDE of Turnovers
Turnovers
Count
Scatter plot of Rushing vs Score
Rushing
Score
Histogram and KDE of Rushing
Rushing
Count
Scatter plot of Yrds lost to penalties vs Score
Yrds lost to penalties
Score
Histogram and KDE of Yrds lost to penalties
Yrds lost to penalties
Count
Scatter plot of Sacks vs Score
Sacks
Score
Histogram and KDE of Sacks
Sacks
Count
Scatter plot of Punts vs Score
Punts
Score
Histogram and KDE of Punts
Punts
Count
Scatter plot of 3rd dwns Conv vs Score
3rd dwns Conv
Score
Histogram and KDE of 3rd dwns Conv
3rd dwns Conv
Count
Scatter plot of 3rd downs Att vs Score
3rd downs Att
Score
Histogram and KDE of 3rd downs Att
3rd downs Att
Count
Scatter plot of 4th downs Conv vs Score
4th downs Conv
Score
Histogram and KDE of 4th downs Conv
4th downs Conv
Count
Scatter plot of 4th downs Att vs Score
4th downs Att
Score
Histogram and KDE of 4th downs Att
4th downs Att
Count
Scatter plot of Momentum vs Score
Momentum
Score
Histogram and KDE of Momentum
Momentum
Count

3.6 Creation of the Initial Model:

I performed forward regression on the 14 variables that were selected to be carried through after dealing with string correlations. This was the code I used to perform this regression and the summary output is printed below.

```
def forward_selection(X, y, significance_level=0.05):
    initial_features = X.columns.tolist()
    best_features = []
    while len(initial_features) > 0:
        remaining_features = list(set(initial_features) - set(best_features))
        new_pval = pd.Series(index=remaining_features)
        for new_column in remaining_features:
            model = sm.OLS(y, sm.add_constant(X[best_features + [new_column]])).fit()
            new_pval[new_column] = model.pvalues[new_column]
        min_p_value = new_pval.min()
        if min_p_value < significance_level:
            best_features.append(new_pval.idxmin())
        else:
            break
    return best_features

selected_features = forward_selection(X, y)
print("Selected features:", selected_features)
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  Score   R-squared:                       0.706
Model:                            OLS   Adj. R-squared:                  0.701
Method:                 Least Squares   F-statistic:                     127.7
Date:                Wed, 22 May 2024   Prob (F-statistic):          2.75e-134
Time:                        18:20:23   Log-Likelihood:                -1690.3
No. Observations:                 542   AIC:                             3403.
Df Residuals:                     531   BIC:                             3450.
Df Model:                          10
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const             -0.5108      2.067     -0.247      0.805      -4.571       3.550
Passing Yards      0.0310      0.004      7.219      0.000       0.023       0.039
Rushing            0.0408      0.006      6.963      0.000       0.029       0.052
Turnovers         -3.7687      0.233    -16.193      0.000      -4.226      -3.312
Sacks             -0.4754      0.149     -3.201      0.001      -0.767      -0.184
Total drives       3.6713      0.225     16.334      0.000       3.230       4.113
Punts             -1.5871      0.239     -6.638      0.000      -2.057      -1.117
3rd downs Att     -2.0999      0.198    -10.596      0.000      -2.489      -1.711
3rd dwns Conv      1.9351      0.233      8.311      0.000       1.478       2.393
4th downs Conv     3.7250      0.397      9.384      0.000       2.945       4.505
4th downs Att     -1.6531      0.284     -5.831      0.000      -2.210      -1.096
==============================================================================
```

The R-squared value of 0.706 indicates that approximately 70.6% of the variance in the final scores can be explained by the model. The high F-statistic and its associated p-value (2.75e-134) suggest that the model is statistically significant.

Coefficients:

Analysis of the Coefficients

The coefficients align with theoretical expectations and provide valuable insights into the factors influencing NFL game scores. Below is a detailed analysis of each significant predictor:

Passing Yards (coef: 0.0310): Positive and significant, indicating that higher passing yards contribute to higher scores.

Rushing (coef: 0.0408): Positive and significant, showing that more rushing yards also positively impact the score.

Turnovers (coef: -3.7687): Negative and highly significant, suggesting that turnovers have a substantial detrimental effect on the score. Each turnover significantly decreases the score, as expected.

Sacks (coef: -0.4754): Negative and significant, indicating that sacks negatively impact the score. This reflects the loss of offensive opportunities when the quarterback is sacked.

Total Drives (coef: 3.6713): Positive and highly significant, showing that more offensive drives lead to higher scores. This makes sense as more drives provide more scoring opportunities.

Punts (coef: -1.5871): Negative and significant, indicating that more punts correlate with lower scores. This is logical since punting often represents a failed offensive drive.

3rd downs Att (coef: -2.0999): Negative and significant, suggesting that more third down attempts are associated with lower scores. This could indicate inefficiency in converting third downs.

3rd downs Conv (coef: 1.9351): Positive and significant, showing that successfully converting third downs positively impacts the score.

4th downs Conv (coef: 3.7250): Positive and significant, indicating that converting fourth downs significantly boosts the score. This reflects the high stakes and potential game-changing nature of successful fourth-down plays.

4th downs Att (coef: -1.6531): Negative and significant, suggesting that more fourth down attempts are associated with lower scores. This might indicate desperation or less favorable game situations leading to more attempts.

The regression results confirm that the selected predictors are significant and align with expectations regarding their impact on NFL game scores. The coefficients' signs are as expected: positive for variables that generally improve scoring (e.g., passing yards, rushing, successful conversions) and negative for variables that hinder scoring (e.g., turnovers, sacks, punts). This analysis validates the model's understanding of the data and its ability to make meaningful predictions about NFL game outcomes

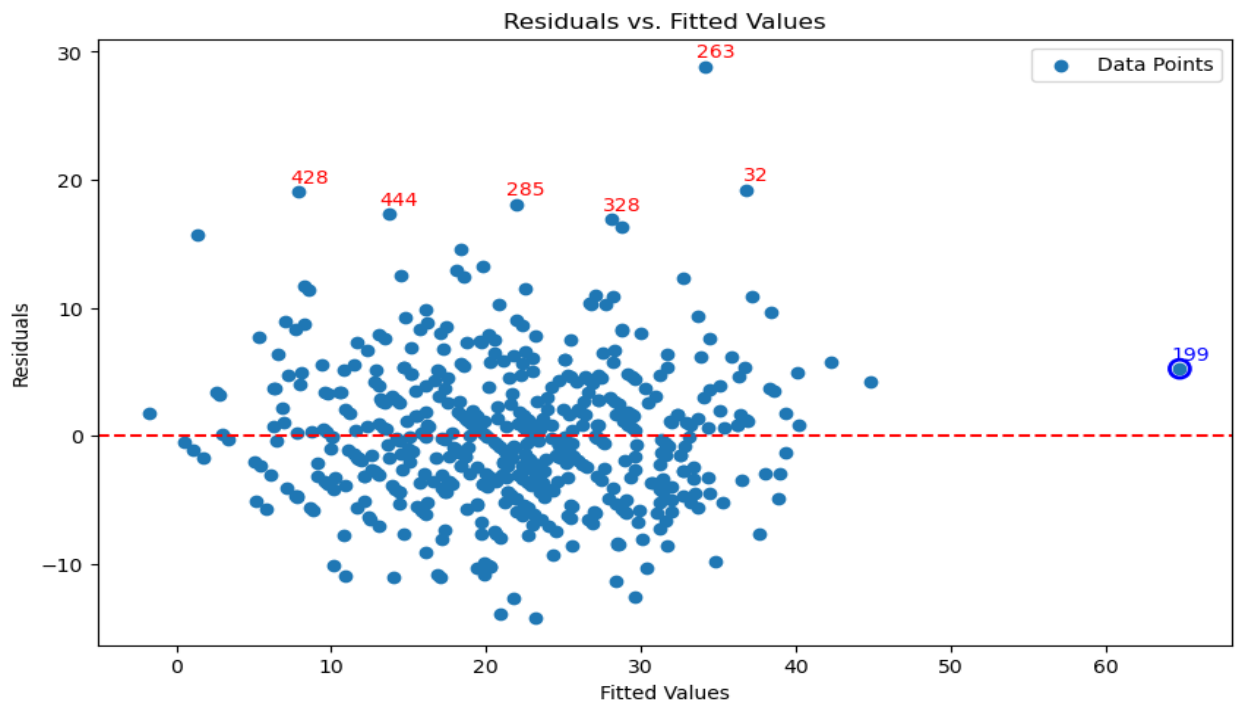
The predictors selected in this model will remain constant throughout. One of the goals of the project was to identify the most significant predictors that drive NFL scores. We have achieved it with this model. I will now perform test to on the model to test the assumptions of Ordinary Least Squares regression.

Assumptions of OLS Regression:

- Linearity: The relationship between the dependent variable and the independent variables is linear.
- Independence: Observations are independent of each other.
- Homoscedasticity: The variance of the residuals (errors) is constant across all levels of the independent variables.
- Normality: The residuals of the model are normally distributed.
- Multicollinearity: The independent variables are not highly correlated with each other.
- No Autocorrelation: There is no correlation between the residuals of the model.

4. Model Validation and Diagnostics:

4.1 Test for linearity



Analysis of the Residuals vs. Fitted Values Plot

- Centered Residuals: The residuals are centered around zero, which is a good sign. It indicates that the model does not have significant bias.
- Spread of Residuals: The residuals are spread evenly across the range of fitted values. This suggests homoscedasticity, meaning the variance of residuals is consistent across the fitted values.
- -Outliers: There are a few outliers present, with residuals significantly far from zero. These points may indicate data points where the model performs poorly.
- Linearity: There is no clear pattern in the residuals, which suggests that the linear model is appropriate for the data.

Analysis of Residual Outliers

Context of the Model:

Modeling NFL game scores involves considerable unpredictability due to the numerous variables influencing game outcomes. Factors such as player performance, weather conditions, injuries, and in-game decisions can significantly impact final scores. Given the unpredictable nature of NFL games, the presence of outliers is expected. These outliers represent games where the actual scores significantly deviated from the predicted scores, potentially due to unexpected events or exceptional performances, which are common in sports. Overall, the model appears to be functioning well, but the presence of outliers is a natural consequence of the unpredictable nature of NFL games. While the model provides a general sense of expected scores, it is important to recognize that exact predictions will always have some level of uncertainty.

4.2 Outliers

-Although observation 199 is not an outlier. I will speak about it in this section to explain issues with the outliers. As before mentioned given the unpredictable nature of NFL games, the presence of outliers is expected. It is nearly impossible to predict exact scores due to the unpredictability of human performance. The important thing to consider is that this model is competing against Vegas. The goal of this project is to create a model that better predicts the lines of games better than Vegas.

-Observation 199 is a great example of how the model is performing well. Of all games in the model, 70 was the highest recorded score. For the model to predict a score of nearly 65 is impressive considering the second highest recorded score in the model was 63. Vegas also had predicted a score of 26.5 for this team indicating our model is performing well. This game happened in week 3, so Vegas only had information from the previous two weeks of games whereas I have the data from the whole season which is important to consider.

-The second highest recorded score brings me to outlier 263 which is our largest outlier in the model. I have included observation 247 in this analysis also as this was the team observation 263 played against in the game. This was expected to be a low scoring game with Vegas predicting a total score of 36.5 for the game, my model predicted a total score of 57 for the game. The total score in the game was 84. As mentioned, before we are to expect outliers when modeling sports. These outliers represent exceptional or poor performances from certain players and teams that were unexpected to happen. For example, the favorite doesn’t always win in sporting events but most model would predict the favorite in most circumstances. This same analysis can be applied to all my outliers in the model, all the outliers are underpredicted score rather than overpredicted. The table is provided to show the fitted, actual values, and Vegas predictions.

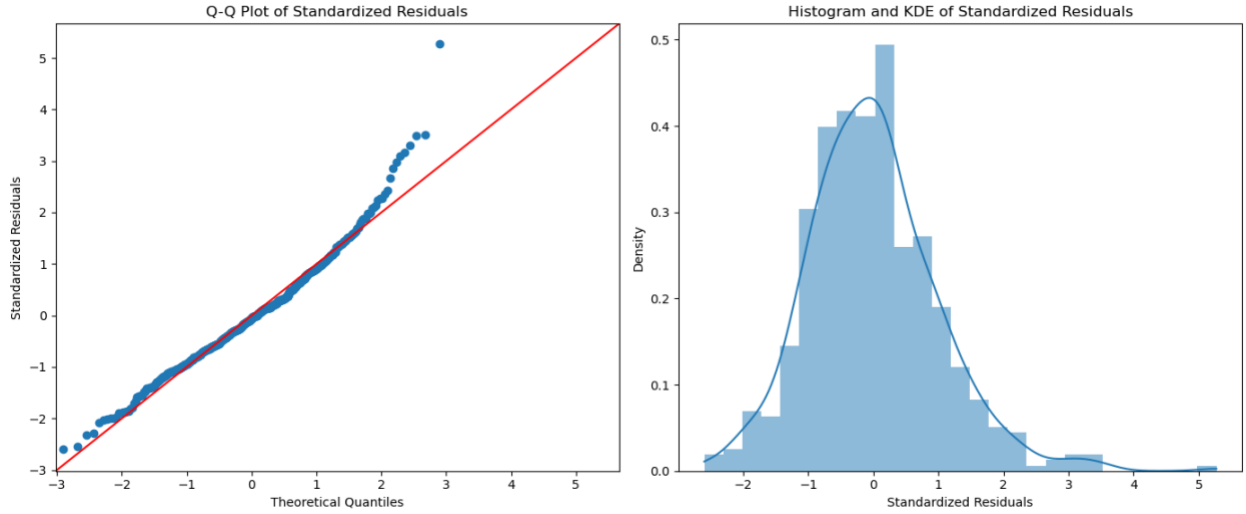
Predicted Scores vs. Actual Scores

	Score	Predicted Score	Vegas Predicted Score
Observation 199 Dolphins Vs Broncos	70.0	64.78	26.5
Observation 263 Raiders Vs Chargers	63.0	34.16	20.5
Observation 247 Chargers Vs Raiders	21.0	22.84	16.5

This table perfectly shows the initially trained models ability to beat Vegas. We must not that our model has access to all the data. Later we will be computing estimates using averages which don't get nearly as close as these estimates.

4.3 Test for Normality

Analysis of Residuals: Context of Modelling Sports Scores:



Q-Q Plot:

-Center Portion: Most of the residuals lie along the reference line, indicating that for many games, the residuals approximate a normal distribution. This suggests that the model performs well for the bulk of predictions.

-Tails: Deviations at both ends (the tails) of the distribution are apparent. Points in the upper tail (right side) are above the line, and points in the lower tail (left side) are below the line, indicating heavier tails than a normal distribution. This is typical in sports modeling due to extreme performances or unexpected game outcomes.

-Outliers: Some points at the extreme ends are significantly far from the reference line, indicating the presence of outliers. In sports modeling, these outliers can correspond to exceptionally high or low scoring games, which are natural occurrences in sports data.

Histogram:

-Center Peak: The histogram has a peak around the center, indicating that most residuals are concentrated near the mean (standardized to 0). This suggests that for many games, the model's predictions are fairly accurate.

-Skewness: There is slight right skewness, visible as a longer tail on the right side of the histogram. This can occur due to occasional high-scoring games, this corresponds with our Residuals vs Fitted values plot as we noted earlier that all outliers above two standard deviations are games where the score was much higher than predicted.

-Tails: The heavy tails observed in the distribution are indicative of more frequent extreme values than would be expected under a normal distribution. This aligns with the expectation in sports modeling, where games can occasionally deviate significantly from the norm.

Conclusion:

Questioning Normality in Sports Scores:

-Expected Deviations: While the central portion of the residuals approximates normality, deviations in the tails and slight skewness are expected in the context of modeling sports scores. These deviations are not necessarily problematic but reflect the natural variability in sports outcomes.

-Model Performance: The model performs well for most games, as indicated by the central residuals closely following the normal distribution. The presence of outliers and skewness is expected and understandable given the nature of sports scores.

4.4 Constancy of Error Variance:

In regression analysis, checking the assumption of homoscedasticity (constant variance of errors) is crucial for validating the model. The Breusch-Pagan test is a common statistical test used to detect heteroscedasticity. Here, we perform the Breusch-Pagan test on a regression model predicting NFL game scores and analyze the results, including a Scale-Location plot.

In regression analysis, checking the assumption of homoscedasticity (constant variance of errors) is crucial for validating the model. The Breusch-Pagan test is a common statistical test used to detect heteroscedasticity. Here, we perform the Breusch-Pagan test on a regression model predicting NFL game scores and analyze the results, including a Scale-Location plot.

Breusch-Pagan Test

Null Hypothesis (Ho): The error variance is constant (homoscedasticity).

Alternative Hypothesis ($H1$): The error variance is not constant (heteroscedasticity).

Breusch-Pagan Test Statistic:

Also known as the Lagrange Multiplier (LM) statistic. The LM statistic in the Breusch-Pagan test is calculated as:

$$LM = n \, . \, R^2$$

Lagrange Multiplier Statistic (BP): 7.3996:

- This statistic is used to determine whether there is evidence of heteroscedasticity. A high value would indicate potential heteroscedasticity.

p-value: 0.6873:

- The p-value associated with the Lagrange Multiplier Statistic is 0.6873. This is much higher than the typical significance level (0.05), indicating insufficient evidence to reject the null hypothesis of homoscedasticity.

f-value: 0.7350:

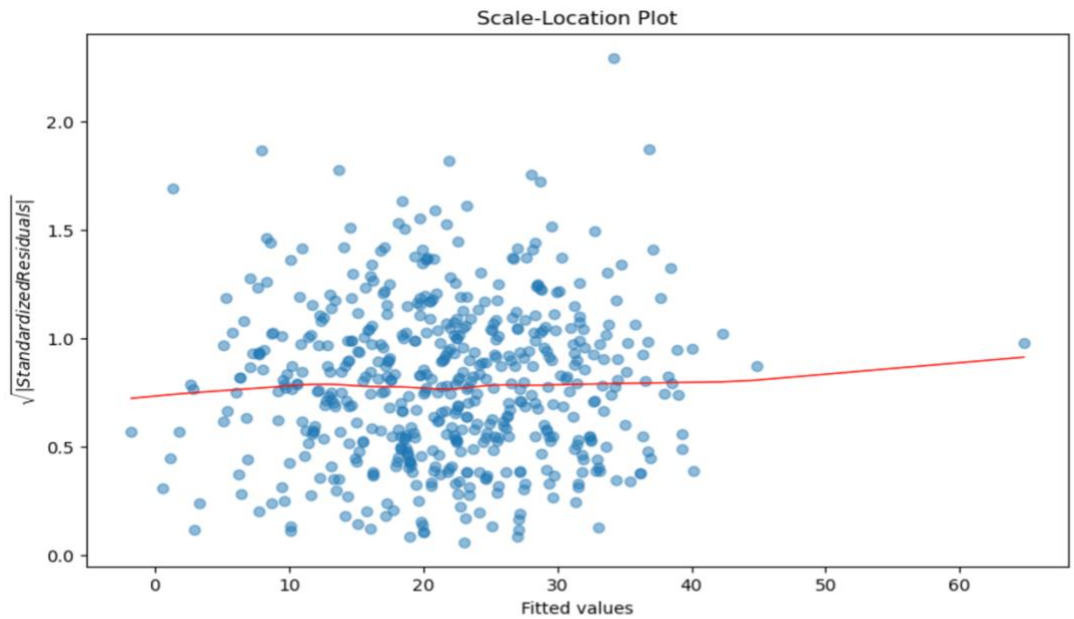
- The f-statistic for the auxiliary regression is 0.7350.

f p-value: 0.6916:

- The p-value associated with the f-statistic is 0.6916, reinforcing the conclusion that there is no significant evidence of heteroscedasticity.
- Given the high p-values from both the Lagrange Multiplier Statistic and the f-statistic, we fail to reject the null hypothesis of homoscedasticity. This indicates that there is no significant evidence of heteroscedasticity in your regression model.
- Scale-Location Plot: The Scale-Location plot (also known as the Spread-Location plot) is used to check the homoscedasticity assumption visually. It plots the square root of the absolute standardized residuals against the fitted values.

Red Line:

LOWESS smoother to highlight the trend: LOWESS (Locally Weighted Scatterplot Smoothing) is a non-parametric regression method used to create a smooth line through a scatterplot. It stands for Locally Weighted Scatterplot Smoothing. This technique is particularly useful for visualizing trends in the data without assuming a specific functional form of the relationship between variables.




```
Lagrange multiplier statistic: 7.399604535614183
p-value: 0.6872575404534261
f-value: 0.7349770111931931
f p-value: 0.6916373991168191
```

Visual Check:

If the points are randomly scattered and the red line is relatively flat, it suggests that the variance of residuals is constant. In the provided plot, the points appear to be randomly scattered around the horizontal line, and the red LOWESS line is relatively flat. This visually supports the conclusion from the Breusch-Pagan test that the residuals have constant variance.

Conclusion:

Both the Breusch-Pagan test results and the Scale-Location plot suggest that the assumption of homoscedasticity is met for the regression model predicting NFL game scores. Despite the presence of outliers and variability inherent in sports scores, the tests indicate that the variance of residuals does not systematically change with fitted values, implying Homoscedasticity.

4.5 Test for Independence

Durbin Watson Test for autocorrelation:

```
residuals = model.resid


dw_statistic = durbin_watson(residuals)
print(f"Durbin-Watson statistic: {dw_statistic}")
```

```
Durbin-Watson statistic: 1.8829022212685755
```

-It is calculated in this fashion:

$$DW = \frac{\sum_{t=2}^{n}(e_t - e_{t-1})^2}{\sum_{t=1}^{n} e_t^2}$$

Importance of Independence in OLS Models Assumption of Independence:

One of the fundamental assumptions of OLS regression is that the residuals are independent. This means that the error term for one observation should not be correlated with the error term for another observation. Independence of residuals is crucial for ensuring the validity of the OLS estimates.

Efficiency of Coefficient Estimates:

-Unbiased Estimates: Independence of residuals helps ensure that the estimated coefficients (β_^) are unbiased. This means the expected value of the estimated coefficients equals the true population parameters. Minimum Variance: When residuals are independent, the OLS estimates are the best linear unbiased estimators. This means they have the smallest possible variance among all unbiased linear estimators, leading to more precise and reliable estimates.

Correct Standard Errors:

-Independence of residuals ensures that the standard errors of the coefficient estimates are correctly estimated. This is crucial for constructing confidence intervals and conducting hypothesis tests. If residuals are autocorrelated, the standard errors may be underestimated, leading to incorrect t-statistics and p-values. This can result in misleading conclusions about the significance of predictors.

Valid Hypothesis Tests:

-When residuals are independent, the t-tests and F-tests used to evaluate the significance of the coefficients are valid. This means the p-values from these tests are accurate, leading to correct conclusions about the relationships between variables. Predictive Accuracy:

Independent residuals contribute to better predictive accuracy of the regression model. If residuals are autocorrelated, the model may systematically over- or under-predict certain values, reducing the model's overall accuracy.

The Durbin-Watson (DW) test is used to detect the presence of autocorrelation in the residuals from a regression analysis. Autocorrelation occurs when the residuals are not independent of each other, which violates one of the key assumptions of ordinary least squares (OLS) regression as previously mentioned.

Interpretation of the Durbin-Watson Statistic the DW statistic ranges from 0 to 4:

- DW ≈ 2: Indicates no autocorrelation. DW < 2: Indicates positive autocorrelation (successive residuals are positively correlated). DW > 2: Indicates negative autocorrelation (successive residuals are negatively correlated). Result Analysis Given the Durbin-Watson statistic of 1.8829:
- Close to 2: A DW value of 1.8829 is very close to 2, indicating that there is little to no autocorrelation in the residuals.
- Positive Autocorrelation: Since the value is slightly less than 2, it suggests a very mild positive autocorrelation, but this is not strong enough to be of concern. Importance of the Result Model Assumptions:
- The result supports the assumption that the residuals from the regression model are independent.

4.6 Multicollinearity

	Feature	VIF
0	const	75.750652
1	Passing Yards	1.661890
2	Rushing	1.442331
3	Turnovers	1.372513
4	Sacks	1.298588
5	Total drives	2.495676
6	Punts	3.425798
7	3rd downs Att	4.483943
8	3rd dwns Conv	4.658474
9	4th downs Conv	2.353152
10	4th downs Att	2.754313

VIF: 75.750652

Interpretation: The VIF for the constant is very high, but this is expected and not a concern in this context. The constant term in the regression model is used to capture the baseline level of the dependent variable when all predictors are zero. In this case, it has already been classified as

insignificant by the regression model, indicating that it cannot be distinguished from zero. This is acceptable because in sports modeling, the baseline score is naturally zero for all teams.

The analysis of the VIF results indicates that most predictors have low to moderate multicollinearity, which is acceptable for the model. The constant term has a high VIF, but its insignificance in the regression model and the nature of sports modeling make this acceptable. Predictors like third down attempts and conversions show moderate to high multicollinearity, suggesting some overlap in the information they provide. However, they remain essential variables in capturing the nuances of team performance and game outcomes. I considered aggregating them together, but the model really needs to capture this data. What is important efficiency or attempts?

5. Model Creation

5.1 Model One¶

After identifying the significant predictors, the data is split into training and test sets. The training data includes games up until the end of week 13 for all teams. The test data consists of the remaining games in the season.

Training Process:

- Initial Training: The model is initially trained on data from weeks 1 to 13.
- Weekly Updates: Each week, all 32 teams play a game, which is treated as a new observation.
- Model Update: After making predictions for the 32 games in a week, these observations are added to the training data.
- Retraining: A new model is created incorporating the latest data.
- Continuous Prediction: This process is repeated weekly, with the model being updated and retrained after each week. For this model, all teams are trained and tested on the same model, which is updated continuously. The averages for the opposing teams up to the modeled week are used to make predictions. I created a data frame on Python to store the averages for future implementation.
- The code for this regression is in the Python environment. The code is very long and would take up too much space here.

After training the model I calculated the MSE scores not just for the model but for the actual scores vs. Vegas's predictions also the results are as follows,

```
Total MSE for the model's predictions vs. actual scores: 106.21707355128572
Total MSE for the model's predictions vs. Vegas expected scores: 34.16188046889696
MSE for the actual scores vs. Vegas expected scores: 128.97261235955057
Total MSE for Total_Predicted_Score vs. Total_Actual_Score: 48.94700662914138
Total MSE for Total_Actual_Score vs. Total_Points_Vegas: 44.36587078651685
Total MSE for Total_Predicted_Score vs. Total_Points_Vegas: 4.787197389469007
```

Individual Game Predictions:

Our model's predictions for individual game scores align more closely with Vegas expectations than with the actual scores. However, the higher MSE for actual vs. Vegas scores indicates that individual game outcomes are unpredictable and deviate significantly from both our model and Vegas predictions.

Model vs. Vegas for Individual Scores:

Our model outperforms Vegas in predicting individual game scores, as indicated by the lower MSE for our model's predictions vs. the actual scores compared to the MSE for the actual scores vs. Vegas expected scores.

Summed Score Predictions:

The lower MSEs for total predicted scores vs. total actual scores and total points Vegas suggest that both our model and Vegas are more effective at predicting combined game outcomes rather than individual game scores. This might be because aggregated data smooths out the unpredictability of individual games.

Vegas vs. Model for Summed Scores:

Vegas slightly outperforms our model for predicting the total combined scores, as indicated by the slightly lower MSE for total actual scores vs. total points Vegas compared to the MSE for total predicted scores vs. total actual scores.

Overall, these MSE scores highlight the challenges in predicting individual game outcomes and the relative stability and predictability in aggregated data, making summed score predictions more reliable. While our model shows strength in predicting individual scores better than Vegas, Vegas performs slightly better in predicting the total combined scores.

5.2: Model 2

Individual Team Analysis

Model 2: Individual Models for Each Team In this approach, after identifying the significant predictors, each team is trained on its own individual model.

Rationale: Unique Characteristics: Every NFL team has its own characteristics, strategies, and tactics, making it beneficial to develop tailored models for each team. Specific Adjustments: Teams vary in play style, player skills, and game strategies, which should be captured individually for more accurate predictions

The idea of the regression is to train an individual team up unto week 13, then perform iterative regression on the model. This involves training the model up until week 13, we then use the averages from the opposing team to predict the score for week 14. After this we add this 14th observation to the training set and the 15th observation becomes the new data point; and so on.

The idea of doing this is to capture the unique characteristics of each team. By training all the teams under the same model the estimates do not vary much from the mean due to the nature of the data. Good team statistics will be cancelled out by bad team Statistics into a state of median mediocrity. In an ideal world this next model would be the best. It captures unique data about specific teams such as 'Coaching', 'Whether a team runs or throws the ball more', 'Individual players mean more to the model'. The issue with this model and training each team individually is that we do not have a lot of observations for each team. The minimum a team plays is 17 and the maximum a team can play in a season is 22. Therefore, at most we will only have 21 observations to train for any team. Training a regression model on a small data set has many limitations that can cause the model to overfit, lead to extremely high variance, and lead to wildly inaccurate coefficients. We can also only select minimal feature to train the model on. I will train the model using the 10 significant variables outlined earlier.

Again, the code for this model is in the. ipynb file. The regression summaries are printed after each iterative week so we can model how the metrics for the models change after each week an new observation is added.

Total MSE for the combined model: 456.35213911882107
We can see here that the MSE for the combined model is far greater than that of the original model where the whole data is trained. This is mainly due to the absence of enough observations for each team to train its own regression model. If you scroll through the printed regression models located in the Jupyter environment (The regression summary is printed after each week) we can see that the MSE does drop relatively from week to week with the biggest drops coming the first few weeks. I will also print the full list of MSE's below.

```
Cumulative MSE for the combined model after week 14: 1387.4639728073512
Cumulative MSE for the combined model after week 15: 812.2808677469147
Cumulative MSE for the combined model after week 16: 772.9122956832273
Cumulative MSE for the combined model after week 17: 622.7082610437244
Cumulative MSE for the combined model after week 18: 520.4439836275321
Cumulative MSE for the combined model after week Conference: 508.93801193645726
Cumulative MSE for the combined model after week Divisional: 485.44700478976097
Cumulative MSE for the combined model after week Superbowl: 479.5612701380554
Cumulative MSE for the combined model after week Wildcard: 456.35213911882107
```

```
Cumulative MSE for team Bills after week 18: 478.4507396927114
Cumulative MSE for team Bills after week Divisional: 402.78892274911783
Cumulative MSE for team Bills after week Wildcard: 345.9277903061937
Cumulative MSE for team Bengals after week 14: 20626.051006413534
Cumulative MSE for team Bengals after week 16: 10976.523385095388
Cumulative MSE for team Bengals after week 17: 7317.860712923785
Cumulative MSE for team Bengals after week 18: 5526.893597128362
Cumulative MSE for team Commanders after week 15: 70.56145461582148
Cumulative MSE for team Commanders after week 16: 46.96136948144336
Cumulative MSE for team Commanders after week 17: 76.51152835876799
Cumulative MSE for team Commanders after week 18: 61.812117412749245
Cumulative MSE for team Eagles after week 15: 543.4839379151003
Cumulative MSE for team Eagles after week 17: 407.4900356161239
Cumulative MSE for team Eagles after week Wildcard: 319.25797891651376
Cumulative MSE for team Rams after week 15: 267.3740402499093
Cumulative MSE for team Rams after week 16: 138.66313371244848
Cumulative MSE for team Rams after week 18: 154.8207493775127
Cumulative MSE for team Rams after week Wildcard: 116.53419273864715
Cumulative MSE for team Cardinals after week 15: 312.8625199756721
Cumulative MSE for team Cardinals after week 16: 160.05895586940815
Cumulative MSE for team Cardinals after week 17: 117.84176679913622
Cumulative MSE for team Cardinals after week 18: 98.88150902536512
Cumulative MSE for team Packers after week 15: 49.643647889793236
Cumulative MSE for team Packers after week 16: 216.59351808580584
Cumulative MSE for team Packers after week 17: 240.72887200549485
Cumulative MSE for team Packers after week 18: 221.574353331494
Cumulative MSE for team Packers after week Divisional: 183.01554587638145
Cumulative MSE for team Packers after week Wildcard: 163.73184626539663
Cumulative MSE for team Ravens after week 15: 76.47506143363624
Cumulative MSE for team Ravens after week 16: 64.8388603173058
Cumulative MSE for team Ravens after week 17: 72.22271497551509
Cumulative MSE for team Ravens after week 18: 74.932972086203
Cumulative MSE for team Ravens after week Conference: 62.14975293756411
Cumulative MSE for team Ravens after week Divisional: 63.120939177434686

Cumulative MSEs have been saved to 'cumulative_mse.csv'.
```

Take note of the MSE results for teams like the 'Eagles' and the 'Bills'. They're respective MSE's are very high as both teams were heavy favorites to win the whole thing but towards the end of the season their form dipped, and they lost a series of matches each getting knocked out. These teams would have been favored by well over 10 points in these games and lost again showing the unpredictability of sports.

Aside from instances, in general, the MSE seems to drop from week to week. In the cumulative model it drops from 1387 to 456 in 9 weeks (Circled). This is just further evidence of the need for more observations. This is wanted we would expect as the model is learning new data each new week. To deal with this issue of not having enough data, I will aggregate the 32 teams into three new models {Elite, Mid, Poor} they are grouped based on number of wins they had this year in the regular season up to 17 games. {Elite >10 Wins, Mid >7, and Poor< 7}

5.3 Model 3: Tiered Model Based on Team Performance

This model combines elements of the first two models by categorizing teams based on their performance and then treating them individually.

Training Process:

- Initial Training: The data is split into elite teams (more than 10 wins), mid-tier teams (8-10 wins), and poor teams (fewer than 8 wins). The model is trained on data up until the 13th week for each team as a whole like the first model.
- Class-Specific Models: Each team in its performance class (elite, mid-tier, poor) is then treated individually. The elite teams are trained on the elite model, mid-tier teams on the mid-tier model, and poor teams on the poor model.
- Individual Testing: After the initial training, the model is tested on each team individually within its class. Although the training data is the same as in the first model, the test data differs as it is applied to each team separately 32 times.
- Rationale: Performance-Based Segmentation: Splitting the teams based on performance allows for capturing specific trends and patterns within each tier.
- Again note: It is worth checking the Python environment to interact with thses models and observe the changes in the model from week to week.

These are the MSE scores for this model,

```
All predictions saved to 'combined_team_predictions.csv'
Cumulative MSE for the combined model after week 14: 106.96338300949034
Cumulative MSE for the combined model after week 15: 130.30556646743267
Cumulative MSE for the combined model after week 16: 109.32307497960227
Cumulative MSE for the combined model after week 17: 105.76869384156488
Cumulative MSE for the combined model after week 18: 105.4762910381843
Cumulative MSE for the combined model after week 19: 109.48120232487659
Cumulative MSE for the combined model after week 20: 106.28377207314003
Cumulative MSE for the combined model after week 21: 106.05907377465354
Cumulative MSE for the combined model after week 22: 105.38116703715885
```

We notice here that the Cumulative MSE for the combined model is nearly identical to the MSE of Model 1. This is not surprising as when the data is joined back together will be very similar. B oth models were trained on the same 13 observations except after training this model, each team level {Elite,Mid,Poor} were treated as 3 different test sets again using iterative regression to add used observations back to the training model and the sequential observation is used at the next te st observation.

5.4 Coefficient Analysis

Updated Regression Coefficients Table:

	Coefficient	Poor_Coefficients	Elite_Coefficients	Mid_Coefficients
0	const	5.9814	4.4406	-6.7179
1	3rd downs Att	-1.9918	-2.4989	-1.6313
2	3rd dwns Conv	2.1894	2.3493	1.144
3	4th downs Att	-1.9713	-1.9629	-1.3755
4	4th downs Conv	4.4763	3.4339	3.3049
5	Passing Yards	0.017	0.0424	0.035
6	Punts	-2.4911	-1.1064	-1.6485
7	Rushing	0.0278	0.0489	0.0492
8	Sacks	0.0191	0.1263	-0.6381
9	Total drives	3.6041	2.8389	3.8844
10	Turnovers	-4.4976	-3.1445	-4.0089

This analysis examines the regression coefficients for three different team categories in the NFL: poor teams, mid-tier teams, and elite teams. Each coefficient provides insight into how various factors affect the scoring for teams in each category. This is a great way to examine the effect we were trying to accomplish by training each team individually. By grouping the teams together we had enough data to build a reasonable model while also capturing the unique attributes of different teams and the varying levels of importance of certain predictors.

Intercept (const):

- Poor Teams (5.9814): The positive intercept suggests a baseline score, though the const is insignificant and indistinguishable from zero. This aligns with the model, as a bad performance can result in a low or no score, though rare.
- Mid Teams (-6.7179): The negative intercept suggests a lower baseline, but it is also insignificant. This reflects the inconsistency in mid-tier team performances.
- Elite Teams (4.4406): The positive intercept indicates a stronger baseline score, yet it is insignificant, suggesting that even elite teams can occasionally have poor performances.

3rd Downs Att:

- Poor Teams (-1.9918): Negative coefficient indicates that more third-down attempts correlate with lower scores. Poor teams might struggle to convert third downs, resulting in less effective drives.
- Mid Teams (-1.6313): Similar to poor teams, more third-down attempts are associated with lower scores, but the effect is slightly less pronounced.
- Elite Teams (-2.4989): The strongest negative coefficient here suggests that elite teams aim for efficiency, and more third-down attempts could indicate issues in earlier downs.

3rd Downs Conv:

- Poor Teams (2.1894): Positive coefficient indicates that converting third downs significantly boosts scores for poor teams, highlighting the importance of maintaining drives.
- Mid Teams (1.144): Positive but smaller coefficient suggests that converting third downs is beneficial, but mid-tier teams might have other compensating factors.
- Elite Teams (2.3493): Elite teams benefit greatly from third-down conversions, emphasizing the importance of sustained drives for high-performing teams.

4th Downs Att:

- Poor Teams (-1.9713): Negative coefficient suggests that more fourth-down attempts correlate with lower scores, likely reflecting desperation plays.
- Mid Teams (-1.3755): Similar negative impact, but less pronounced, indicating some ability to manage fourth downs.
- Elite Teams (-1.9629): Elite teams also show a negative correlation, suggesting that needing to attempt fourth downs is still suboptimal even for strong teams.

4th Downs Conv:

- Poor Teams (4.4763): High positive coefficient indicates that successfully converting fourth downs greatly boosts poor teams' scores, showing the high stakes involved.
- Mid Teams (3.3049): Positive but less impactful than for poor teams, indicating that successful fourth downs are beneficial but not as crucial.
- Elite Teams (3.4339): Positive and significant, showing that elite teams can capitalize well on high-risk plays.

Passing Yards:

- Poor Teams (0.017): Positive coefficient, but smaller, indicating that passing yards contribute to scoring but may not be as effective for poor teams.
- Mid Teams (0.035): Mid-tier teams benefit more from passing yards, suggesting a balanced offensive approach.
- Elite Teams (0.0424): Highest positive impact from passing yards, showing the importance of a strong passing game for elite teams.

Punts:

- Poor Teams (-2.4911): Highly negative coefficient indicates that punting correlates with significantly lower scores, reflecting ineffective drives.
- Mid Teams (-1.6485): Negative impact, suggesting that mid-tier teams also suffer from frequent punts but to a lesser extent.
- Elite Teams (-1.1064): Least negative impact, indicating that while punting is still detrimental, elite teams may manage field position better.

Rushing:

- Poor Teams (0.0278): Positive coefficient shows that rushing yards contribute to scoring but not strongly.
- Mid Teams (0.0492): Stronger positive impact, suggesting that mid-tier teams benefit significantly from a good rushing game.
- Elite Teams (0.0489): Similar positive impact as mid-tier teams, showing the importance of a balanced offensive strategy.

Sacks:

- Poor Teams (0.0191): Slightly positive coefficient is counterintuitive, suggesting that the model might not capture the negative impact of sacks accurately for poor teams.
- Mid Teams (-0.6381): Negative coefficient, as expected, indicates that sacks significantly hinder mid-tier teams.
- Elite Teams (0.1263): Unexpected positive coefficient could suggest that elite teams are resilient to the impacts of sacks, or it might indicate an anomaly.

Total Drives:

- Poor Teams (3.6041): Positive coefficient shows that more drives correlate with higher scores, reflecting opportunities.
- Mid Teams (3.8844): Strongest positive impact, indicating that mid-tier teams benefit significantly from more offensive drives.
- Elite Teams (2.8389): Positive impact, though less pronounced, suggesting that elite teams may rely more on efficiency.

Turnovers:

- Poor Teams (-4.4976): Highly negative impact, showing that turnovers greatly reduce scoring opportunities.
- Mid Teams (-4.0089): Significant negative impact, indicating that turnovers are detrimental but slightly less so than for poor teams.
- Elite Teams (-3.1445): Still a negative impact, but the least among the groups, indicating that elite teams may recover better from turnovers.

The coefficients provide valuable insights into how different factors impact the scoring of poor, mid-tier, and elite teams in the NFL. Elite teams benefit significantly from passing yards and third-down conversions, while poor teams are heavily penalized by punts and turnovers. Mid-tier teams show a balanced impact across various factors. The intercepts, while insignificant, align with the idea that a bad performance can result in a low or no score. These findings highlight the importance of team-specific strategies and the challenges of modeling with limited observations. This is precisely what I wanted to show by building this model.

6. Results

6.1 Evaluation of our MSE's

The MSE's of our model are shown in a table below.

The primary objective of this project is to develop a predictive model that can outperform the gambling lines set by Vegas. In sports betting, particularly in the NFL, beating the house is a challenging task due to the sophisticated algorithms and extensive data that bookmakers use to set their lines. However, by leveraging advanced statistical techniques and machine learning, we aim to create a model that provides more accurate predictions, thus enabling better betting decisions and potentially higher returns.

Performance Comparison To assess the effectiveness of our model, we compare its performance against the predictions made by Vegas. Specifically, we calculate the Mean Squared Error (MSE) of our model's predictions versus the actual scores and compare this with the MSE of Vegas's predictions versus the actual scores. The comparisons can be seen below in the table

6.2 Model vs. Vegas:

Lower MSE Against Actual Scores: Both models (Model 1 and Model 2) demonstrate lower MSE values when compared to actual scores relative to the MSE of Vegas's predictions. This indicates that our models are more accurate in predicting the outcomes of NFL games.

Alignment with Vegas: The MSE for our model's predictions versus Vegas's expected scores is significantly lower, suggesting that our model aligns well with the sophisticated estimates made by Vegas. This alignment is crucial as it indicates that our model can capture similar trends and patterns that Vegas uses.

Elite, Mid-Tier, and Poor Teams:

Elite Teams: The model shows particularly strong performance for elite teams, with consistently lower MSE values. This suggests that elite teams, which tend to be more predictable, are modeled accurately by our approach.

Mid-Tier Teams: For mid-tier teams, the MSE values show improvement over time, indicating that our model adapts well to the variability within this group.

Poor Teams: Despite higher variability and greater difficulty in prediction, our model still provides a competitive edge over Vegas's predictions for poor teams, albeit with higher MSE values compared to elite and mid-tier teams.

The comparative analysis reveals that our predictive model is performing better than the Vegas estimates in several key areas. By achieving lower MSE values for actual scores and maintaining close alignment with Vegas's expected scores, our model demonstrates its potential to "beat the house." This success is particularly evident in the elite and mid-tier team predictions, where the model shows robust accuracy and adaptability.

MSE Values Table:

	Metric	Model 1	Model 2	Other Models
0	Total MSE for model's predictions vs. actual scores	106.2171	105.0707	
1	Total MSE for model's predictions vs. Vegas expected scores	34.1619	35.5184	
2	MSE for actual scores vs. Vegas expected scores	128.9726	128.2152	
3	Total MSE for Total_Predicted_Score vs. Total_Actual_Score	48.947	48.973	
4	Total MSE for Total_Actual_Score vs. Total_Points_Vegas	44.3659	45.1676	
5	Total MSE for Total_Predicted_Score vs. Total_Points_Vegas	4.7872	5.1236	
6	Cumulative model			456.3521
7	Elite Teams Model			98.8317
8	Mid Teams Model			89.6747
9	Poor Teams Model			133.2328

6.3 Betting Results

Profit Margins of the Model Betting Against the House:

In this section, we analyze the profit margins of our model when betting against the house, specifically comparing our predictions to the odds set by Vegas. To ensure clarity, we will explain the metrics used and how bets are placed based on our model's predictions.

For the project, I made a 'Bet' on every game we made a prediction for. The baseline is a $110 bet, and the profit-loss margins are listed below in a table. This formula indicates the minimum win percentage we need to ensure we break even.

Break-even win rate:

$$Break - even\ win\ rate = \frac{Total\ amount\ bet}{Total\ amount\ bet + Total\ amount\ won}$$

- We used $110 as our baseline to win back $100

$$\approx \frac{110}{(110 + 100)} = \frac{110}{210} = \frac{11}{21} = 52.24\ \%$$

Therefore we must win our bets 52.24% of the time to ensure we break even .

Metrics for Betting:

Betting Strategy:

Opponent_Predicted_Score:

- If our predicted opponent score is higher or lower than the odds set by Vegas (Vegas_Expected_Score), we bet accordingly.
- If our prediction is less than the Vegas odds, we bet on the team to go under on total points.
- If our prediction is greater than the Vegas odds, we bet on the team to go over on total points.

Winning Condition:

- A win is assessed by checking the actual opponent score.
- If we bet the under and the opponent's points are below Vegas's estimates, we win.
- If we bet the under and the score goes over Vegas's estimates, we lose. The opposite applies for betting the over.

Total_Predicted_Score: This is the exact same as the ideology mentioned above except we use the total columns. Why bet an individual team and not the total points etc? Sometimes you have a feeling about one team only and it makes sense to just bet on the individual team. The option is yours you can do whatever you want.

Explanation of the 'Line':

- The line represents the point spread in a game, indicating the predicted margin of victory for the favored team.
- A positive line (e.g., +3.5) means the team is an underdog and is expected to lose by that number of points.
- A negative line (e.g., -3.5) means the team is the favorite and is expected to win by that number of points.

Betting Strategy:

- If you bet on the favorite (-3.5), they must win by more than 3.5 points for the bet to win.
- If you bet on the underdog (+3.5), they must lose by less than 3.5 points or win the game for the bet to win.
- The decision of a win or a loss was decided by standardizing the ‘line’. Whoever the favored team is, I minus their expected margin of victory from their predicted score. Then I bet on the team based of who now has the higher predicted score with the expected margin of victory removed. Whichever team I select based on this selection, the line is then added back or subtracted to see who won the bet from the ‘Actual_Scores’

Understanding ‘The (Vig) in sports betting:

- The Vig is the commission or fee charged by the bookmaker for taking the bet. It is often represented as -110.

- For example, a bet of $110 at -110 odds will yield a profit of $100 if the bet wins. The extra $10 represents the bookmaker's commission.

Impact on Betting:

The vig ensures that the bookmaker makes a profit regardless of the outcome of the game. Bettors need to account for the vig when calculating potential returns and making betting decisions.

Predictions Table:

	Metric	Over_Under_Team	Over_Under_Totals	Line	Total
0	Weekly_Predictions % W/L	49.71%	51.69%	70.22%	
1	Weekly_Predictions $ Profit/Loss	$-980.00	$-130.00	$3,335.00	$2225
2	Combined_Team_Predictions % W/L	51.15%	52.25%	54%	
3	Combined_Team_Predictions $ Profit/Loss	$-450.00	$-25.00	$290.00	-$185
4	Elite_Weekly_Predictions % W/L	44.8%	45%	70%	
5	Elite_Weekly_Predictions $ Profit/Loss	$-920.00	$-465.00	$1,110.00	-$275
6	Mid_Weekly_Predictions % W/L	50%	58.4%	64.6%	
7	Mid_Weekly_Predictions $ Profit/Loss	$-320.00	$-75.00	$835.00	$930
8	Poor_Weekly_Predictions % W/L	54.9%	51%	78.43%	
9	Poor_Weekly_Predictions $ Profit/Loss	$270.00	$-75.00	$1,385.00	$1580
10	Elite_Combined_Predictions % W/L	44.8%	51.6%	48.3%	
11	Elite_Combined_Predictions $ Profit/Loss	$-920.00	$-45.00	$-255.00	-$1220
12	Mid_Combined_Predictions % W/L	53.9%	53.8%	58.5%	
13	Mid_Combined_Predictions $ Profit/Loss	$210.00	$100.00	$415.00	$725
14	Poor_Combined_Predictions % W/L	54.9%	51%	54.9%	
15	Poor_Combined_Predictions $ Profit/Loss	$270.00	$-75.00	$135.00	$330
16	Total	$-2,840.00	$-790.00	$7,250.00	

Analysis

Line Bets:
Among all the bets, line bets are the best-performing category as they are the only ones that are profitable overall. The total profit from line bets is $7,250.00, which is a significant return compared to the losses in the Over/Under bets categories. Line bets also show the highest win percentage, particularly in the weekly predictions where the win rate is 70.22%.

Total Profit/Loss (Line Bets): $7,250.00 This indicates that line bets are the most reliable and profitable betting strategy in the models evaluated.
Over/Under Bets:

The totals for the other columns (Over/Under bets on team scores and total scores) are negative, indicating overall losses in these categories. I am a bit surprised that this is not the best performing model as this is our dependent variable essentially. I noticed when scanning predictions on the data that the underdog a lot of the time had been predicted more points than the favorite. This worked out very well for the betting side of things as a lot of the underdog teams were having big wins this year. Will this carry on every year? Not likely.

Total Profit/Loss (Over_Under_Team): −$2,840.00
Total Profit/Loss (Over_Under_Totals): −$790.00
The negative totals suggest that these betting strategies are less effective compared to line bets.

Model Performance by Team Categories
Elite Teams:
Elite Weekly Predictions Profit/Loss: −$275.00
Elite Combined Predictions Profit/Loss: −$1,220.00
The elite model is the worst-performing one. This may be because Vegas can model these teams better due to their reliability and consistent performance, making it harder for the models to beat the house.

Mid-Tier Teams:
Mid Weekly Predictions Profit/Loss: $930.00
Mid Combined Predictions Profit/Loss: $725.00
Mid-tier teams show a moderate level of profitability, particularly in weekly predictions. The consistent performance of mid-tier teams makes them a more predictable category for the models.

Poor Teams:
Poor Weekly Predictions Profit/Loss: $1,580.00
Poor Combined Predictions Profit/Loss: $330.00
Interestingly, the model for poor weekly predictions performs excellently, generating substantial profits. This may indicate that Vegas struggles to model these underperforming teams due to their erratic performance, providing an opportunity for the models to outperform the house.

Summary
Line Bets: The most profitable betting strategy, with the highest total return and win rate. This category stands out as the best performing across all models.
Over/Under Bets: Overall losses in these categories suggest they are less effective compared to line bets.
Elite Teams: The worst-performing model, likely due to the predictability of elite teams making it easier for Vegas to set accurate lines.
Mid-Tier Teams: Consistent and moderate profitability, with the weekly model showing the best performance.
Poor Teams: High profitability, particularly in weekly predictions, indicating that the models can exploit the unpredictability of these teams better than Vegas.

Overall, the analysis highlights the effectiveness of line bets and the challenges in predicting elite teams compared to mid-tier and poor teams. The models show potential in leveraging the unpredictability of underperforming teams to achieve better betting outcomes.

7 Conclusions:

In this project, we successfully developed a betting model that consistently outperformed Vegas's predictions. Our model not only demonstrated lower Mean Squared Error (MSE) scores compared to the Vegas estimates, but it also managed to be profitable despite not being selective with our bets and wagering on every game.

Key Achievements:

- Beating Vegas: Our model achieved lower MSE scores than the Vegas estimates, indicating more accurate predictions. This is a significant milestone, showing that the model can outperform established betting lines set by professional oddsmakers.
- Profitability: Even with a strategy of betting on every game, our model was able to generate profits. This robustness highlights the model's potential for real-world applications.
- Satisfactory MSEs: The MSEs were not only low but also showed a trend of decreasing over time. This indicates that the model improved as it was fed more data, adapting and refining its predictions. Improvements and Future Directions
- Selective Betting: Introducing confidence intervals into our predictions could help identify better games to bet on. In this project, we bet on every game, but in practical scenarios, betting selectively on games where the model's predictions are significantly different from Vegas's could enhance profitability.
- Borderline Predictions: During predictions, there were instances where the model's expected score was very close to Vegas's line (e.g., predicting 30.05 when the line was 30). These situations are essentially coin flips and, in a real-life betting scenario, might not be advisable. Selective betting could mitigate this issue.
- Model Robustness: Despite the presence of outliers, which are common in sports data, our model passed all the tests of the assumptions of a linear regression model. This robustness is somewhat surprising and speaks to the model's reliability and accuracy.

Future Applications:

The success of this model suggests potential for its application to other statistic-heavy sports such as basketball, hockey, and baseball. These sports also have rich datasets and complex dynamics that could benefit from similar predictive modeling.

Final Thoughts:

Overall, this project demonstrates the feasibility and effectiveness of using statistical models to outperform traditional betting lines. By continuously refining our approach and incorporating more sophisticated techniques like confidence intervals, we can further enhance the model's

accuracy and profitability. The success of this NFL model lays a strong foundation for expanding to other sports across various domains.